

vk_deeplearning_models - One-Page App Summary

What it is

A compact Python app that trains a feedforward neural network on MNIST digits. It uses stochastic gradient descent with backpropagation and sigmoid activations.

Who it's for

Primary persona: ML learners who want a minimal, readable neural-net training example.

What it does

- Loads MNIST data from ../data/mnist.pkl.gz using gzip + pickle.
- Reshapes each image into a 784x1 input vector.
- One-hot encodes training labels into 10x1 target vectors.
- Builds a dense network from a layer-size list (example: [784, 20, 10]).
- Initializes biases and weights with Gaussian random values.
- Trains with mini-batch SGD, backprop, and configurable epochs/batch size/eta.
- Evaluates test predictions each epoch and prints accuracy counts.

How it works (repo-evidence architecture)

- run_network1.py orchestrates: load data, build Network, call SGD.
- mnist_loader.py handles data IO and preprocessing for train/val/test splits.
- network1.py contains model state, feedforward, SGD loop, backprop, evaluate.
- Data flow: MNIST file -> wrapper transforms -> SGD mini-batches -> epoch metrics.

How to run (minimal steps)

1. Create Python env and install numpy (dependency files: Not found in repo).
2. Place dataset at ../data/mnist.pkl.gz relative to this repo root.
3. Run: python3 run_network1.py

Not found in repo

- README, packaged data file, and dataset download/source instructions.
- CLI/config system, tests, and model checkpoint export path.